Session 1 (ASK): Solutions and facilitation notes

Foundations of Clinical Research · Level 1 · Neudata · *Instructor copy. Do not share before the session.*

These are **model** answers. Many other answers are acceptable: reward work that is specific, consistent and could be acted on. The course is online and the whole class works as one group: use the "what good looks like" boxes when you read the chat and when you type the class answer on the shared screen.

Quick quiz (segment 1.1): Rank the evidence (poll)

Answer (weakest to strongest): A (expert opinion), then B (registry cohort), then C (randomised trial, CLASSIC), then D (systematic review / meta-analysis of trials). Poll option 1: "A, B, C, D".

Source	Design	Real example
A	Expert opinion	None
B	Observational cohort	Similar to Paper 2: Hyun 2023, *Ann Intensive Care*. Positive cumulative fluid balance on ICU days 2–3 associated with higher 28-day mortality (day 2 cohort HR 2.13, 95% CI 1.48–3.06)
C	Randomised controlled trial	CLASSIC: Meyhoff et al., *N Engl J Med* 2022. 1,554 ICU adults with septic shock; 90-day mortality 42.3% restrictive vs 42.1% standard
D	Systematic review / meta-analysis	Pooled analyses of fluid-restriction trials

Discussion points (chat)

- What could make C weaker than B? Very small size, many losses to follow-up, poor allocation concealment, an outcome that does not matter to patients, or a population unlike ours.
- Why might B and C disagree? In a cohort, sicker patients receive more fluid, so fluid can look harmful even if it is not. This is **confounding by indication** (taught in Session 4). Randomisation removes this.
- The pyramid ranks **designs**, not individual papers. Quality must still be checked (Session 8).

Check your understanding (segment 1.1): types of research (chat)

Item	Answer
A Counting patients with septic shock who are fluid-positive on day 3 this year	Quantitative, primary
B Interviewing ICU nurses about why they hesitate to stop IV fluids	Qualitative, primary
C Re-analysing 3 years of ICU records for fluid balance and death	Quantitative, **secondary** (data already exist; still needs ethics review)

Item	Answer
D Survey of fluid prescribing + focus groups with doctors	Mixed methods, primary

Check your understanding (segment 1.2): spot the PICO in Paper 1's question (poll + chat)

"In adults with septic shock in the first 24 hours of vasopressors, does restricting non-resuscitative fluids, compared with usual care, reduce cumulative fluid balance at day 5?"

Poll answer: C, cumulative fluid balance, is the outcome. D (day 5) is the time point; A (septic shock) is part of the population; B is the intervention.

P	I	C	O	T
Adults with septic shock, in the first 24 hours of vasopressors	Restricting non-resuscitative fluids (maintenance fluids and fluids for drug dilution)	Usual care	Cumulative fluid balance (ml/kg)	Day 5

Common error: choosing "septic shock" as the outcome. It has already happened and defines who is eligible (P).

Exercise 1 (whole class): rescue the running example

Use the version the class chose in the poll. If the vote is split, work on A. Hypotheses are taught in segment 1.3 (Exercise 2), so the class does not write H0 and H1 here.

Version A: pilot trial (experimental, PICO)

P	Adults with septic shock in our ICU, within 24 h of starting vasopressors
I	Restrictive fluid protocol
C	Usual care
O	Cumulative fluid balance (ml/kg)
T	Day 5

In adults with septic shock in our ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?

Type: analytical, experimental (a pilot randomised trial, like Paper 1).

FINER: F green/amber (about 50 patients over 12–18 months) · I green · N amber to green (local setting, different protocol) · E amber (deferred or surrogate consent, Session 7) · R green.

Version B: cohort (observational, PECO)

P	Adults with sepsis or septic shock admitted to our ICU
E	Positive cumulative fluid balance by day 3 (Exposure: we do not assign it)
C	Neutral or negative cumulative fluid balance by day 3
O	Death from any cause
T	By day 28

In adults with sepsis in our ICU, is a positive cumulative fluid balance by day 3, compared with a neutral or negative balance, associated with higher 28-day mortality?

Type: analytical, observational. This is Paper 2's question and the main question of the FLUID-ICU course study (Session 5).

FINER watch-points: F depends on patient numbers and on records of daily fluid balance; confounding by indication (sicker patients get more fluid) must be handled (Session 4).

Version C: descriptive (P + O)

P	Adults with septic shock admitted to our ICU over 6 months
O	Positive cumulative fluid balance
T	On ICU day 3

What proportion of adults with septic shock in our ICU have a positive cumulative fluid balance on day 3?

Type: descriptive: we **estimate** a proportion with its 95% CI. In segment 1.3 this is the example of a question that needs no hypothesis.

FINER: F green (routine charts) · N amber (local data are new) · E green (usually low risk; still needs approval for records) · R green (shows whether the problem exists here).

What good looks like (Exercise 1)

- All boxes filled; the comparison is named, not "nothing".
- One primary outcome, **measurable**, with a **time point**.
- The sentence reads naturally and contains every PICO element.
- Common faults to correct: two outcomes in one question ("mortality and length of stay"); outcome without time; population too broad ("all patients"); intervention not defined ("better care").

Exercise 2 (individual): aim, objectives and hypothesis for the pilot

- *Aim:* To compare a restrictive fluid protocol with usual care in adults with septic shock in our ICU.
- *Objectives:* Primary: to compare cumulative fluid balance (ml/kg) at day 5 between groups. Secondary: to estimate the recruitment rate and protocol adherence (feasibility). (Optional third: to describe 28-day mortality and kidney injury needing dialysis in each group, exploratory.)

- *Hypothesis:* H0: cumulative fluid balance at day 5 is the same in both groups. H1: it differs.
- *Descriptive version C (no hypothesis):* To estimate the proportion of adults with septic shock in our ICU who have a positive cumulative fluid balance on day 3.

Vague-verb check (exit question 2): rewrite "To understand fluid balance in our ICU" as *"To estimate the proportion of adults with septic shock in our ICU who have a positive cumulative fluid balance on day 3."*

What good looks like

- H0 always states "no difference / no association".
- H1 is usually written as "differs" (in either direction), even when we expect benefit, because the intervention could also harm. One- vs two-sided tests are taught in Session 6.
- The hypothesis uses the **same** outcome and time point as the primary objective.
- Correct the misconception: data can **reject** H0 or **fail to reject** it; we never "prove H0". "No significant difference" is not "no difference" (Paper 1, n = 48, could not detect a mortality difference because it was not powered for it).

Exercise 3 (individual): Proposal builder 1, what good looks like

The trainer models Part 1 on screen for the pilot version (A) of the running example; each participant then writes Part 1 for their own question. Use this model as the quality benchmark, not as the answer. The references, problem statement and title are added next Saturday (Session 2).

PICO(T): P adults with septic shock within 24 h of starting vasopressors · I restrictive fluid protocol · C usual care · O cumulative fluid balance (ml/kg) · T at day 5.

Question: *In adults with septic shock in our ICU, does a restrictive fluid protocol, compared with usual care, reduce cumulative fluid balance at day 5?*

FINER: F green/amber (pilot of about 50 patients over 12–18 months) · I green · N amber to green (local setting, different protocol) · E amber (deferred or surrogate consent, Session 7) · R green.

Aim: To compare a restrictive fluid protocol with usual care in adults with septic shock in our ICU.

Objectives:

1. *(Primary)* To compare cumulative fluid balance (ml/kg) at day 5 between groups.
2. To estimate recruitment rate and protocol adherence (feasibility).
3. To describe 28-day mortality and kidney injury needing dialysis in each group (exploratory).

Hypothesis: H0: cumulative fluid balance at day 5 is the same in both groups. H1: it differs.

Feedback guide when reading drafts (not a grade)

Element	Good	Needs work
Topic	One focused problem from the participant's department	Three problems merged; "improve care in the ICU"
Question	All PICO(T) elements; one primary outcome with time point	Two outcomes; no comparison; vague population
FINER	Each letter justified; red letters fixed	Ticked without thought

Element	Good	Needs work
Aim/objectives	One aim; 2–3 measurable objectives; primary marked	"To understand…", "to study…"; more than 4 objectives
Hypothesis	Matches primary objective; H0 and H1; or correctly "descriptive"	Hypothesis about a different outcome

Exercise 4: Synonym list (homework)

Model for the running example: **P** septic shock · sepsis · vasopressor-dependent sepsis; **I** fluid restriction · restrictive fluid · conservative fluid · de-resuscitation; **O** mortality · death · survival.

Exit question: answers

1. **Feasible.** A mortality outcome needs well over a thousand patients (Novel is also amber because large trials already exist).
2. e.g. *"To estimate the proportion of adults with septic shock in our ICU who have a positive cumulative fluid balance on day 3."*

Extra practice (optional, at home): model answers

A. Scenarios: model PICO(T) answers

Scenario A: Ventilator-associated pneumonia (experimental). P adults on invasive mechanical ventilation for ≥ 48 h in the ICU · I twice-daily nurse checklist for head-of-bed elevation 30–45° · C usual care (no checklist) · O ventilator-associated pneumonia (stated definition) · T up to ICU day 28.

In adults ventilated for at least 48 hours in the ICU, does a twice-daily nurse checklist for head-of-bed elevation, compared with usual care, reduce the incidence of ventilator-associated pneumonia by ICU day 28? FINER watch-point: **F** (VAP definitions vary and VAP is not very common).

Scenario B: Pressure injuries (descriptive first). *What is the incidence of hospital-acquired pressure injuries (stage 2 or worse) among adults admitted to our medical ward over 6 months?* P adults admitted to the medical ward · O new pressure injury, stage 2 or worse · T during the admission, over 6 months. Descriptive: no hypothesis; estimate incidence with a 95% CI. Later analytical question: *does 2-hourly vs 4-hourly repositioning reduce new pressure injuries in at-risk adults?*

Scenario C: Delirium (experimental). P adults aged 65+ after hip-fracture surgery · I daily multicomponent non-drug delirium-prevention bundle · C usual care · O incident delirium (validated tool, e.g. 4AT or CAM) · T first 7 postoperative days. FINER watch-points: **E** (consent from patients who may become confused) and **F** (staff time).

Scenario D: Stroke door-to-needle time (observational, PECO). P adults with acute ischaemic stroke receiving thrombolysis in our emergency department · E pre-hospital ambulance alert · C no pre-alert · O door-to-needle time (minutes) · T from arrival to start of thrombolysis. The outcome is continuous, so the hypothesis is about a difference in median or mean time.

Scenario E: Medication errors (experimental or before–after). P adults transferred from ICU to a general ward · I pharmacist-led medication reconciliation at transfer · C usual transfer process · O patients with at least one medication discrepancy · T within 48 hours of transfer. Watch-point: define "medication error"; one primary outcome only.

B. FINER: model answers

#	Main problem	Why
1	**Feasible**	Whole country, 10 years, mortality: far beyond any department's resources. (Large vitamin D trials already exist too, so Novel also fails.)
2	**Ethical**	Withholding antibiotics in septic shock causes harm; there is no equipoise. An ethics committee would refuse it.
3	**Novel**	The answer is already measured monthly. This is an audit question rather than a research question, so use the existing data.
4	**Relevant** (and Interesting)	No plausible mechanism and no impact on practice.

These are also on an extra slide ("More practice: which FINER letter fails?").

C. Hypotheses

a) Early mobilisation. H0: ICU length of stay is the **same** with early mobilisation and with usual care. H1: ICU length of stay **differs** between early mobilisation and usual care.

b) Proportion with positive fluid balance. No hypothesis needed. This is descriptive: we *estimate* a proportion with its 95% confidence interval.

Extra slides: research, audit or QI? (answers)

Project	Answer	Ethics route
Checking whether fluid balance is charted daily, against the unit protocol	**Audit**	Local/clinical-governance approval; usually no ethics committee
Testing a daily fluid-balance checklist in PDSA cycles	**Quality improvement**	Local approval; ethics review if it adds risk or goes beyond routine care
Randomising patients to restrictive or usual fluids	**Research**	Ethics committee approval + trial registration before the first patient
Interviewing nurses about why they hesitate to stop fluids	**Research** (qualitative)	Ethics committee approval (consent, confidentiality)
Reviewing 3 years of ICU records to see if fluid balance is linked to mortality	**Research** (secondary data, observational)	Ethics committee approval or a formal waiver of consent

Key point: rules differ between institutions. When in doubt, ask the ethics committee **before** starting; journals ask for approval or a waiver at submission.